A COMPLETE PROOF CANDIDATE FOR CARLET'S CONJECTURE
RECIPROCAL COMPACTIFICATIONS, QUADRATIC COMPLETE INTERSECTIONS, AND EXPLICIT BETTI BOUNDS

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Abstract. This paper organizes a complete proof candidate for the zero-sum subspace form of Carlet's conjecture. For odd n, the target is that for every $3 \le k \le n-3$ there exists a k-dimensional $\mathbb{F}_2$-subspace $E \subset \mathbb{F}_{2^n}$ such that $\sum_{x \in E \setminus \{0\}} x^{-1} = 0$. The proposed route converts the zero-sum condition into the vanishing of the X^2 coefficient of a subspace polynomial, realizes it as a Cartier section on a Pink–Schieder reciprocal compactification, constructs an explicit quadratic model for the full-rank stratum, and derives point-count estimates from stratified cohomology and finite certificates. Three explicit certificates for $(n,k) = (29,13), (29,14), (31,14)$ have been independently checked by exact computation. The general reciprocal-geometric argument in Sections 4–10 has not received a complete independent line-by-line review; accordingly, the main statement is presented as a proof candidate rather than as a certified theorem.

Manuscript status. The three finite-field certificates have been independently checked exactly. This does not certify the general geometric argument. The main proof candidate remains subject to independent specialist review and has not been formalized in Lean.

Revision note (November 2026)

This release removes the earlier word-processing cover material and presents the manuscript with a journal-style mathematical front matter. It adds clarifications on (i) defining the formal product $L_j(T)$ over an arbitrary $\mathbb{F}_2$-algebra before using any linear-independence conclusion; (ii) the vanishing range of compactly supported cohomology used in the total Betti bound; and (iii) the Frobenius-compatible identification of the local intersection complex with the constant sheaf, including the Tate twist in duality. The complete arguments are recorded in the Chinese edition's corresponding sections; this English edition's source proof text follows the supplied English candidate manuscript.

On 5 October 2026, Kaimin Cheng posted the preprint *Counting zero-sum subspaces for the multiplicative inverse function* (arXiv:2610.06445), whose abstract claims a proof of the full conjecture by a counting and Fourier-analytic method. We have not audited that proof and make no priority determination. Its existence is relevant to the current literature and is recorded here without assessing its correctness.

Keywords: Carlet conjecture; finite fields; multiplicative inverse; subspace polynomials; reciprocal planes; intersection cohomology; Betti bounds.

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Supplementary mathematical clarifications

1. The reciprocal polynomial over an arbitrary base ring. Let A be any commutative $\mathbb{F}_2$-algebra and let $x_1, \dots, x_j \in A$. Define the formal polynomial

$$L_j(T) = \prod_{\epsilon \in \mathbb{F}_2^j} \left(T + \sum_{i=1}^{j} \epsilon_i x_i \right) \in A[T].$$

This product and its recursion are defined before any assertion that the x_i are linearly independent. In the reverse construction of Theorem 4.2, the unit conditions first imply that every nonzero binary linear combination is a unit; only then does the tuple define a point of the Drinfeld open.

2. The full degree range in the compact-support Betti sum. The variety X_r is smooth, affine, and pure of dimension $d = r - 2$. Artin vanishing gives $H^j(X_r) = 0$ for $j > d$, and Poincaré duality gives $H_c^i(X_r) = 0$ for $i < d$. The dimension bound gives $H_c^i(X_r) = 0$ for $i > 2d$. Thus the sum in Lemma 7.1 includes every potentially nonzero compactly supported cohomology group; the case $r = 2$ has $d = 0$ and is covered as well.

3. Frobenius structure and duality. The identification of the local intersection complex with the shifted constant sheaf must be taken compatibly with the structure over $\mathbb{F}_2$, hence with Frobenius. For a proper rationally smooth m-dimensional variety, the duality used in (6.2) is $H^i \cong H^{2m-i,\vee}(-m)$, with its Tate twist retained. These clarifications make explicit the Frobenius representation used by the trace formula.

Related work. Kaimin Cheng's preprint, submitted to arXiv on 5 October 2026, claims a proof of the full conjecture by a counting recurrence and Fourier analysis (arXiv:2610.06445v1). We have not audited that proof and make no priority determination.

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1 Introduction and main proof candidate

1.1 Carlet's conjecture

Let $\mathbb{F}_{2^n}$ denote the binary finite field with 2^n elements, and define

$$f_{\mathrm{inv}}(0) = 0, \qquad f_{\mathrm{inv}}(x) = x^{-1} \quad (x \neq 0).$$

Carlet introduced k-th order sum-freedom and showed that, for the multiplicative inverse in the range relevant here, failure of k-th order sum-freedom can be expressed by a zero reciprocal sum on a k-dimensional linear subspace.

Conjecture 1.1 (Carlet)**.** *If n is odd, then for every integer $3 \leq k \leq n-3$ there exists a k-dimensional $\mathbb{F}_2$-subspace $E \subset \mathbb{F}_{2^n}$ such that*

$$\sum_{0 \neq x \in E} x^{-1} = 0.$$

Let K_n denote the set of dimensions for which such a subspace exists. Carlet–Hou and subsequent work give complementary-dimension symmetry,

$$k \in K_n \iff n - k \in K_n.$$

Hence, for odd n, it is enough to treat $3 \leq k \leq (n-1)/2$.

1.2 Status of the results in this manuscript

Theorem 1.2 (Verified finite cases)**.** *Carlet's conjecture holds for $n = 29$ and $n = 31$. More precisely, Appendix A gives explicit zero-sum subspaces for $(29, 13)$, $(29, 14)$, and $(31, 14)$; known results and complementary-dimension symmetry then cover all admissible dimensions.*

The finite certificates have been checked in two independent exact ways: by constructing the full subspace polynomial and verifying that its X^2 coefficient is zero, and by exhaustive summation of the inverses of all nonzero elements in the subspace. The supplementary script `verify_carlet_paper.py` reproduces these checks.

Main Proof Candidate 1.3. *For every odd n and every $3 \leq k \leq n-3$, there exists a k-dimensional $\mathbb{F}_2$-subspace $E \subset \mathbb{F}_{2^n}$ such that*

$$\sum_{0 \neq x \in E} x^{-1} = 0.$$

Status: independent audit pending. Sections 4–10 give a complete derivation of Candidate 1.3. Any conclusion depending on the new reciprocal-geometric bridge should be read as part of the proof candidate until independently checked.

1.3 Strategy

- Write the zero-sum condition as a linear equation $\ell_k = 0$ in reciprocal coordinates and define an ample Cartier divisor D_k inside the Pink–Schieder compactification Q_k.
- Identify the full-rank stratum X_r with an explicit quadratic affine scheme Y_r at the level of schemes; Y_r is a smooth set-theoretic complete intersection.
- Use the Pink–Schieder rank stratification and local product structure, together with Elias–Proudfoot–Wakefield intersection cohomology of reciprocal planes, to prove rational smoothness of Q_k and D_k.
- Use Pink–Schieder strange morphisms to identify the Frobenius-equivariant ℓ-adic cohomology of Q_k with projective space; then use Wan–Zhang weak Lefschetz and Verdier duality to isolate a single unknown middle degree for D_k.
- Bound that middle contribution using the Wan–Zhang Betti estimate for affine complete intersections, and control the lower-rank boundary by an elementary degree bound and Schwartz–Zippel.
- Prove positivity in the central range and close the finite tail using published small-dimensional results and the explicit $n = 29, 31$ certificates.

2 Preliminaries: subspace polynomials and complementary dimensions

2.1 Subspace polynomials

Definition 2.1. *For a finite $\mathbb{F}_2$-subspace E, define*

$$L_E(X) = \prod_{u \in E} (X + u).$$

If $\dim_{\mathbb{F}_2} E = r$, then L_E is a monic linearized polynomial of the form

$$L_E(X) = c_0 X + c_1 X^2 + c_2 X^4 + \cdots + c_r X^{2^r}, \qquad c_0 \neq 0, \quad c_r = 1.$$

Lemma 2.2 (Reciprocal-sum criterion)**.** *Let $E \neq \{0\}$. Then*

$$\sum_{0 \neq u \in E} u^{-1} = \frac{c_1}{c_0}.$$

In particular, E is zero-sum if and only if $c_1 = 0$.

Proof. Set

$$Q_E(X) = \frac{L_E(X)}{X} = \prod_{0 \neq u \in E} (X + u).$$

Since $Q_E(0) = \prod_{0 \neq u \in E} u = c_0 \neq 0$, logarithmic differentiation gives

$$\frac{Q_E'(0)}{Q_E(0)} = \sum_{0 \neq u \in E} u^{-1}.$$

On the other hand, $Q_E(X) = c_0 + c_1 X + c_2 X^3 + \cdots$, so $Q_E'(0) = c_1$. □

Lemma 2.3 (Extension recurrence)**.** *If $v \notin E$ and $E^+ = E \oplus \langle v \rangle$, then*

$$L_{E^+}(X) = L_E(X)^2 + L_E(v)L_E(X).$$

Proof. Because $E^+ = E \sqcup (E + v)$,

$$L_{E^+}(X) = L_E(X)L_E(X + v).$$

Since L_E is linearized, $L_E(X + v) = L_E(X) + L_E(v)$. □

Corollary 2.4 (Last-step linear solver)**.** *If $L_E(X) = c_0X + c_1X^2 + \cdots$ with $c_1 \neq 0$, then $E \oplus \langle v \rangle$ is zero-sum if and only if*

$$L_E(v) = \frac{c_0^2}{c_1}.$$

Proof. By Lemma 2.3, the X^2 coefficient after extension is

$$c_1^+ = c_0^2 + L_E(v)c_1.$$

Now apply Lemma 2.2. Since L_E is $\mathbb{F}_2$-linear, this is an ordinary binary linear system. □

2.2 Complementary dimensions

We use the published complementary-dimension theorem of Carlet–Hou/Hou–Zhao as an external input:

$$k \in K_n \iff n - k \in K_n.$$

Thus, for odd n, it is enough to prove the result for

$$3 \leq k \leq \frac{n-1}{2},$$

equivalently $n \geq 2k + 1$.

3 Reciprocal compactification and the Carlet divisor

3.1 The Pink–Schieder reciprocal ring

Let $V = \mathbb{F}_2^k$. Inside the fraction field of $S_V = \mathrm{Sym}(V)$, Pink–Schieder define

$$R_V = \mathbb{F}_2\big[1/v : 0 \neq v \in V\big].$$

Pink–Schieder assign degree -1 to $1/v$. We reverse the grading so that the reciprocal generators have degree $+1$, and set

$$Q_V = \mathrm{Proj}\, R_V.$$

Pink–Schieder [7, Thm. 1.7, 1.11] prove that R_V is a Cohen–Macaulay normal integral domain and that Q_V is integral, Cohen–Macaulay, and projectively normal. The open stratum Ω_V identifies with the Drinfeld period domain $\mathbb{P}^{k-1}$ minus all $\mathbb{F}_2$-rational hyperplanes.

3.2 The Carlet linear section

For $0 \neq v \in V$, write $z_v = 1/v$ and define

$$\ell_k = \sum_{0 \neq v \in V} z_v, \qquad D_k = V(\ell_k) \subset Q_V, \qquad X_k = D_k \cap \Omega_V.$$

Lemma 3.1. *The element ℓ_k is nonzero in R_V. Hence D_k is an effective ample Cartier divisor cut out by a nonzero section of $\mathcal{O}(1)$, and $\dim D_k = k - 2$.*

Proof. Write the last coordinate as $T = x_k$ and the others as $x' = (x_1, \ldots, x_{k-1})$. Over $K = \mathbb{F}_2(x_1, \ldots, x_{k-1})$ let

$$L(T) = \prod_{u \in \mathbb{F}_2^{k-1}} (T + u \cdot x').$$

This is a linearized polynomial, so its ordinary derivative $L'(T) = C$ is its linear coefficient,

$$C = \prod_{0 \neq u \in \mathbb{F}_2^{k-1}} u \cdot x' \neq 0.$$

Hence

$$\sum_{u \in \mathbb{F}_2^{k-1}} \frac{1}{T + u \cdot x'} = \frac{L'(T)}{L(T)} = \frac{C}{L(T)}.$$

Grouping ℓ_k according to the last coordinate gives

$$\ell_k(x', T) = \ell_{k-1}(x') + \frac{C}{L(T)}.$$

The second term is a nonconstant rational function of T, while the first is independent of T, so the sum cannot vanish identically. Thus $\ell_k \neq 0$. Since R_V is a domain, this section is a nonzerodivisor. With the reversed grading, R_V is generated in degree one, so $\mathcal{O}(1)$ is ample. Therefore D_k is an effective ample Cartier divisor and $\dim D_k = k - 2$. □

Proposition 3.2 (Rational points and zero-sum subspaces)**.** *For $q = 2^n$,*

$$X_k(\mathbb{F}_q) \neq \varnothing$$

if and only if there exists a k-dimensional $\mathbb{F}_2$-subspace $E \subset \mathbb{F}_q$ such that

$$\sum_{0 \neq x \in E} x^{-1} = 0.$$

Proof. A point of $\Omega_k(\mathbb{F}_q)$ may be written $[x_1 : \cdots : x_k]$ and avoids every $\mathbb{F}_2$-rational hyperplane exactly when $x_1, \ldots, x_k$ are $\mathbb{F}_2$-linearly independent. Let $E = \langle x_1, \ldots, x_k \rangle_{\mathbb{F}_2}$. Then $E \setminus \{0\}$ is exactly the set of all $v \cdot x$ with $0 \neq v \in \mathbb{F}_2^k$, and therefore

$$\ell_k(x) = \sum_{0 \neq v \in \mathbb{F}_2^k} (v \cdot x)^{-1} = \sum_{0 \neq y \in E} y^{-1}.$$

Projective rescaling replaces E by λE and multiplies the reciprocal sum by λ^{-1}, so vanishing is preserved. □

4 An explicit quadratic complete-intersection model

4.1 Recursive coordinates

Because Ω_r removes the coordinate hyperplane $x_1 = 0$, the whole open set lies in the affine chart $x_1 \neq 0$. We normalize $x_1 = 1$. Put

$$E_j = \langle x_1, \dots, x_j \rangle, \qquad L_j = L_{E_j}, \qquad L_0(X) = X,$$

and set $\delta_1 = 1$ and $\delta_j = L_{j-1}(x_j)$ for $j \geq 2$. If $C_j = [X]L_j$, Lemma 2.3 gives

$$L_j(X) = L_{j-1}(X)^2 + \delta_j L_{j-1}(X), \qquad C_j = \delta_j C_{j-1}. \tag{4.1}$$

Define

$$f_j = \sum_{u \in E_{j-1}} \frac{1}{x_j + u} = \frac{C_{j-1}}{\delta_j}. \tag{4.2}$$

Then

$$\frac{f_j}{f_{j+1}} = \frac{\delta_{j+1}}{\delta_j^2}. \tag{4.3}$$

Partitioning $\mathbb{F}_2^r \setminus \{0\}$ by the last nonzero coordinate gives

$$\ell_r = f_1 + \cdots + f_r, \qquad f_1 = 1. \tag{4.4}$$

4.2 The quadratic scheme Y_r

Definition 4.1. *Let $Y_r \subset \mathbb{A}^{N_r}_{\mathbb{F}_2}$ have variables*

$$u_{a,j} \ (2 \leq a \leq r, \ 1 \leq j < a), \qquad t_i, d_i \ (2 \leq i \leq r), \qquad c_j \ (1 \leq j < r),$$

with conventions $u_{a,a} = 1$ and $t_1 = 1$. Impose the equations

$$\begin{aligned} u_{a,j}^2 + u_{a,j} &= c_j u_{a,j+1}, \\ t_i d_i &= 1, \\ c_1 t_2 &= 1, \\ c_j t_{j+1} &= t_j \quad (2 \leq j < r), \\ 1 + t_2 + \cdots + t_r &= 0. \end{aligned}$$

The number of variables and equations is

$$N_r = \binom{r}{2} + 3(r-1), \qquad R_r = \binom{r}{2} + 2(r-1) + 1,$$

so $N_r - R_r = r - 2$.

Theorem 4.2 (Explicit quadratic model)**.** *For every $r \geq 2$, the full-rank Carlet stratum X_r is scheme-theoretically isomorphic over $\mathbb{F}_2$ to Y_r.*

Status: key new lemma; independent audit requested.

Proof. **Step 1: construct $X_r \to Y_r$.** For a normalized basis $x_1 = 1, x_2, \dots, x_r$ on X_r, define

$$t_j = f_j, \qquad d_j = t_j^{-1}, \qquad c_j = \frac{t_j}{t_{j+1}}, \qquad u_{a,j} = \frac{L_{j-1}(x_a)}{\delta_j}. \tag{4.5}$$

Every denominator is a unit. Indeed, the definition of Ω_r makes every nonzero $\mathbb{F}_2$-linear combination invertible, hence

$$\delta_j = \prod_{u \in E_{j-1}} (x_j + u)$$

is a unit. Likewise $C_{j-1} = \prod_{0 \neq u \in E_{j-1}} u$ is a unit, so $f_j = C_{j-1}/\delta_j$ is a unit.

Using (4.1), for $a > j$ we obtain

$$u_{a,j}^2 + u_{a,j} = \frac{L_j(x_a)}{\delta_j^2} = \frac{\delta_{j+1}}{\delta_j^2} u_{a,j+1} = c_j u_{a,j+1}.$$

The inverse and ratio equations follow directly from the definitions, while the final zero-sum equation follows from (4.4) and $\ell_r = 0$. Hence we obtain a morphism $\Phi : X_r \to Y_r$.

Step 2: construct $Y_r \to X_r$. Let A be any $\mathbb{F}_2$-algebra and take an A-valued point of Y_r. Set $x_1 = 1$ and $x_a = u_{a,1}$. Define recursively

$$\delta_1 = 1, \qquad \delta_{j+1} = c_j \delta_j^2.$$

We claim that, for every $a \geq j$,

$$L_{j-1}(x_a) = \delta_j u_{a,j}. \tag{4.6}$$

For $j = 1$ this is exactly $x_a = u_{a,1}$. If it holds for j, then

$$L_j(x_a) = \delta_j^2 (u_{a,j}^2 + u_{a,j}) = \delta_j^2 c_j u_{a,j+1} = \delta_{j+1} u_{a,j+1},$$

so induction proves (4.6). Taking $a = j$ gives $L_{j-1}(x_j) = \delta_j$ because $u_{j,j} = 1$.

Step 3: the reverse construction lands in the Drinfeld open. From $t_i d_i = 1$, all t_i are units. The ratio equations give $c_1 = t_2^{-1}$ and $c_j = t_j t_{j+1}^{-1}$, so every c_j and hence every δ_j is a unit. Since

$$\delta_j = L_{j-1}(x_j) = \prod_{u \in E_{j-1}} (x_j + u),$$

each factor $x_j + u$ is a unit. Every nonzero $\mathbb{F}_2$-linear combination has a largest nonzero index j and is of this form. Thus the reverse construction satisfies the functor-of-points condition defining Ω_r.

Step 4: the reverse point satisfies the Carlet equation. If $C_j = [X]L_j$, then $C_j = \delta_j C_{j-1}$. By (4.2), the reconstructed reciprocal functions satisfy

$$\frac{f_j}{f_{j+1}} = \frac{\delta_{j+1}}{\delta_j^2} = c_j = \frac{t_j}{t_{j+1}}.$$

Since $f_1 = t_1 = 1$, induction gives $f_j = t_j$ for all j. Therefore $1 + t_2 + \cdots + t_r = 0$ is equivalent to $\ell_r = 0$.

Step 5: the maps are inverse. Starting from X_r, the reverse map recovers $x_a = u_{a,1} = L_0(x_a)/\delta_1 = x_a$. Starting from Y_r, equation (4.6) recovers every $u_{a,j}$, and the previous paragraph recovers t_j, d_j, c_j. Hence Φ and Ψ are inverse morphisms. □

Corollary 4.3. *The scheme X_r is smooth and affine of pure dimension $r - 2$. More precisely, if*

$$T_r = \{(t_2, \dots, t_r) \in \mathbb{G}_m^{r-1} : 1 + t_2 + \cdots + t_r = 0\},$$

then $X_r \cong Y_r \to T_r$ is finite étale of degree $2^{\binom{r}{2}}$.

Proof. The equations $t_i d_i = 1$ and the ratio equations eliminate all d_i and c_j, leaving T_r as the base. It is a linear hyperplane inside a torus and is smooth of dimension $r-2$. For fixed t, each equation $u_{a,j}^2 + u_{a,j} = c_j u_{a,j+1}$ is monic quadratic in $u_{a,j}$ with derivative 1. Recursing from $j = a-1$ down to 1 gives an iterated finite étale quadratic extension. There are $\sum_{a=2}^{r}(a-1) = \binom{r}{2}$ such steps. □

5 Rank stratification and rational smoothness

5.1 Pink–Schieder rank stratification

Pink–Schieder [7, Thm. 8.2] decompose Q_V as the set-theoretic disjoint union of strata Ω_W indexed by nonzero subspaces $W \le V$, with closure relation

$$\overline{\Omega_W} = \bigcup_{0 \neq W' \le W} \Omega_{W'}.$$

Their extension-by-zero interpretation also shows that, scheme-theoretically, z_v restricts to zero on Ω_W whenever $v \notin W$.

Proposition 5.1. *The Carlet divisor has the stratification*

$$D_k = \bigsqcup_{r=2}^{k} \bigsqcup_{\substack{W \le \mathbb{F}_2^k \\ \dim W = r}} X_r.$$

Proof. On Ω_W, extension by zero identifies the restriction of ℓ_k with ℓ_W, so $D_k \cap \Omega_W = X_{\dim W}$. There are no rank-one points because $\ell_1 = 1/x \neq 0$. The closure relation gives a closed filtration by rank. □

5.2 Local intersection cohomology of the reciprocal normal cone

Pink–Schieder [7, Prop. 8.3] give a stronger local product. If $V = W \oplus V''$, then a Zariski neighborhood of Ω_W in Q_V is locally identified near $\Omega_W \times \{0\}$ with

$$\Omega_W \times \widetilde{Q}_{V''}, \qquad \widetilde{Q}_{V''} = \operatorname{Spec} R_{V''}.$$

Lemma 5.2. *For $\ell \neq 2$, the affine reciprocal cone $\widetilde{Q}_U$ has trivial ℓ-adic local intersection cohomology at its vertex.*

Proof. View $U \setminus \{0\}$ as the collection of nonzero linear forms on U^*. The ring R_U is the corresponding Orlik–Terao/reciprocal-plane coordinate ring. Its matroid is the binary projective geometry $PG(\dim U - 1, 2)$, whose flat lattice is the lattice of subspaces of U. This lattice is modular, and every interval is modular. Elias–Proudfoot–Wakefield [8, Prop. 2.14] prove that $P_M(t) = 1$ if and only if $L(M)$ is modular. Their reciprocal-plane interpretation of the Kazhdan–Lusztig polynomial, together with the local-slice statement [8, Lem. 3.8], identifies the local intersection cohomology at the cone point with the corresponding reciprocal slice. Thus the local intersection cohomology has exactly the one-dimensional contribution expected at a smooth point. □

Proposition 5.3. *Q_k is ℓ-adically rationally smooth.*

Status: combined argument; independent audit requested.

Proof. Take $x \in \Omega_W$. By [7, Prop. 8.3], a neighborhood of x is locally $\Omega_W \times \widetilde{Q}_{V''}$ near $(x, 0)$. The first factor is smooth, while Lemma 5.2 gives trivial local intersection cohomology for the normal factor at its vertex. Smooth-product functoriality of the intersection complex gives the local intersection cohomology of affine space of the same dimension. □

Lemma 5.4 (Smooth slicing)**.** *Let $p : U \to N$ be smooth at x with relative dimension d, and let $h \in \Gamma(U, \mathcal{O}_U)$. If dh is nonzero in the tangent direction of the fiber of p at x, then $V(h) \to N$ is smooth at x of relative dimension $d - 1$. Consequently, étale-locally,*

$$(V(h), x) \cong (N, p(x)) \times \mathbb{A}^{d-1}.$$

Proof. The nonzero relative differential implies that $(p, h) : U \to N \times \mathbb{A}^1$ is smooth at x of relative dimension $d - 1$. Taking the fiber over 0 gives the first statement. Any smooth morphism is étale-locally an affine-space projection. □

Theorem 5.5. *D_k is ℓ-adically rationally smooth.*

Status: key new theorem; independent audit requested.

Proof. Take $x \in D_k$ lying on a rank-r stratum $X_r \subset \Omega_W$. Choose a complement V''. By [7, Prop. 8.3], near x we may write Q_k as $\Omega_W \times N$ with $N = \widetilde{Q}_{V''}$. The restriction of the Carlet section to Ω_W is ℓ_W. By Corollary 4.3, $X_r = \{\ell_W = 0\} \subset \Omega_W$ is smooth of codimension one. Hence $d\ell_W$ is nonzero in an Ω_W-tangent direction. Lemma 5.4 gives

$$(D_k, x) \cong_{\text{ét}} (N, 0) \times \mathbb{A}^{r-2}.$$

Lemma 5.2 and smooth-product functoriality now give the local intersection cohomology of a smooth $(k - 2)$-dimensional space. □

6 Cohomology: universal homeomorphism, weak Lefschetz, and duality

6.1 Frobenius-equivariant cohomology of Q_k

Proposition 6.1. *For every $\ell \neq 2$, the ℓ-adic étale cohomology of Q_k is isomorphic, as a Frobenius representation, to that of $\mathbb{P}^{k-1}$.*

Proof. Pink–Schieder [7, Thm. 9.3] construct morphisms $Q_V \to \mathbb{P}(V^*)$ and $\mathbb{P}(V^*) \to Q_V$ whose composites are powers of Frobenius. They state that the maps are finite, bijective on underlying sets, and radicial, hence finite universal homeomorphisms. By topological invariance of the étale site under universal homeomorphisms [10, Prop. 59.45.4, Tag 03SI], pullback induces an equivalence of étale topoi and therefore an isomorphism on étale cohomology. Since the maps are defined over $\mathbb{F}_2$, this isomorphism commutes with geometric Frobenius. □

6.2 Weak Lefschetz

Put $d = \dim Q_k = k - 1$ and $m = \dim D_k = k - 2 = d - 1$. By Proposition 5.3 and Theorem 5.5,

$$\mathrm{IC}_{Q_k} \cong \mathbb{Q}_\ell[d], \qquad \mathrm{IC}_{D_k} \cong \mathbb{Q}_\ell[m].$$

We use Wan–Zhang [9, Lem. 2.3.1]: if Y is projective, $\mathcal{F} \in {}^pD^{\geq 0}(Y)$, and X is the intersection of r ample Cartier divisors, then

$$H^{j-r}(Y;\mathcal{F}) \to H^{j-r}(X;\mathcal{F})$$

is an isomorphism for $j < 0$ and injective for $j = 0$.

Theorem 6.2. *The restriction*

$$H^i(Q_k, \mathbb{Q}_\ell) \longrightarrow H^i(D_k, \mathbb{Q}_\ell)$$

is an isomorphism for $i < m$ and injective for $i = m$.

Proof. By Lemma 3.1, D_k is an ample Cartier divisor. Take $\mathcal{F} = \mathrm{IC}_{Q_k} = \mathbb{Q}_\ell[d]$, which is perverse. Wan–Zhang gives

$$H^{j-1}(Q_k;\mathbb{Q}_\ell[d]) \to H^{j-1}(D_k;\mathbb{Q}_\ell[d]).$$

Both sides are ordinary cohomology in degree $i = j - 1 + d = j + m$. Thus $j < 0$ is exactly $i < m$, and $j = 0$ is exactly $i = m$. □

6.3 High degrees from duality on D_k

Audit correction

Weak Lefschetz does not imply $H^i(D_k) \cong H^i(Q_k)$ for $i > m$. The inclusion $\mathbb{P}^2 \subset \mathbb{P}^3$ is already a counterexample. The correct argument uses weak Lefschetz below the middle degree and Verdier/Poincaré duality on D_k above it.

Theorem 6.3. *For every $i \neq m$, there is a Frobenius-equivariant isomorphism*

$$H^i(D_k, \mathbb{Q}_\ell) \cong H^i(\mathbb{P}^m, \mathbb{Q}_\ell).$$

Proof. For $i < m$, Theorem 6.2 and Proposition 6.1 give

$$H^i(D_k) \cong H^i(Q_k) \cong H^i(\mathbb{P}^{m+1}),$$

which agrees with $H^i(\mathbb{P}^m)$ in this range. Since D_k is proper and rationally smooth, its intersection complex is the shifted constant sheaf and is Verdier self-dual. Hence

$$H^i(D_k) \cong H^{2m-i}(D_k)^\vee(-m).$$

If $i > m$, then $2m - i < m$, so the right-hand side is already known, including its Tate twist. Projective space satisfies the same duality. □

6.4 Point-count formula

Define the Frobenius-stable quotient

$$V_k = H^m(D_k, \mathbb{Q}_\ell) \Big/ \operatorname{im}\big(H^m(Q_k, \mathbb{Q}_\ell) \to H^m(D_k, \mathbb{Q}_\ell)\big), \qquad b_k = \dim V_k.$$

Corollary 6.4. *For $q = 2^n$,*

$$\#D_k(\mathbb{F}_q) = 1 + q + \cdots + q^m + (-1)^m \operatorname{Tr}(F_q \mid V_k).$$

Proof. Outside the middle degree, Theorem 6.3 gives exactly the Tate cohomology of $\mathbb{P}^m$. If m is even, $H^m(Q_k) = \mathbb{Q}_\ell(-m/2)$ injects by Theorem 6.2 and supplies the middle Tate class of $\mathbb{P}^m$; if m is odd, this term is absent. The remaining quotient is V_k. Apply the Grothendieck–Lefschetz trace formula to the proper scheme D_k. □

Lemma 6.5.

$$|\operatorname{Tr}(F_q \mid V_k)| \le b_k q^{m/2}.$$

Proof. Deligne's general weight theorem gives Frobenius weights at most m on H^m_c. Since D_k is proper, $H^m_c = H^m$. The quotient V_k is Frobenius-stable, so every eigenvalue on V_k has absolute value at most $q^{m/2}$. □

7 An explicit bound for the middle Betti number

7.1 Wan–Zhang affine complete-intersection bound

We use Wan–Zhang [9, Cor. 4.2.4(ii)]: if $V \subset \mathbb{A}^N$ is cut out by R equations of degree at most d and $\dim V = N - R$, then for $0 \le j \le N - R$,

$$\dim H^{N-R+j}_c(V, \mathbb{Q}_\ell) \le \binom{N-j-1}{R-1} d^{N-j}.$$

By Theorem 4.2 and Corollary 4.3,

$$\dim H^{r-2+j}_c(X_r) \le \binom{N_r-j-1}{R_r-1} 2^{N_r-j}, \qquad 0 \le j \le r-2.$$

Lemma 7.1. *The total compactly supported Betti number satisfies*

$$B_c(X_r) < A_r, \qquad A_r = 2^{N_r+1}\binom{N_r-1}{r-2}.$$

Proof. Since $R_r = N_r - (r-2)$,

$$\binom{N_r-j-1}{R_r-1} = \binom{N_r-j-1}{r-2-j} \le \binom{N_r-1}{r-2}.$$

Sum over j and use $\sum_{j\ge0} 2^{-j} < 2$. □

7.2 Rank filtration

By Proposition 5.1 and the closure relation in [7, Thm. 8.2], the union of strata of rank at most r gives a closed filtration. Let

$$\begin{bmatrix} k \\ r \end{bmatrix}_2$$

denote the Gaussian binomial coefficient. The associated compact-support spectral sequence gives

$$B(D_k) = B_c(D_k) \le \sum_{r=2}^{k} \begin{bmatrix} k \\ r \end{bmatrix}_2 B_c(X_r). \tag{7.1}$$

Lemma 7.2 (Gaussian bound)**.** *For $0 \le r \le k$,*

$$\begin{bmatrix} k \\ r \end{bmatrix}_2 < 4 \cdot 2^{r(k-r)}.$$

Proof. Write

$$\begin{bmatrix} k \\ r \end{bmatrix}_2 = 2^{r(k-r)} \prod_{i=0}^{r-1} \frac{1-2^{i-k}}{1-2^{i-r}}.$$

The numerator is less than 1, and

$$\prod_{j \ge 1} (1-2^{-j})^{-1} < 4.$$

□

Lemma 7.3. *If $r = k - s$, then*

$$N_k - N_{k-s} - (k-s)s = \frac{s(s+5)}{2}.$$

Proof. Use $N_t = t(t-1)/2 + 3(t-1) = (t^2+5t-6)/2$ and expand. □

Theorem 7.4 (Closed Betti bound)**.** *For $k \ge 3$,*

$$b_k < \beta_k, \qquad \beta_k = 2^{N_k+2} \binom{N_k - 1}{k-2}.$$

Proof. Since $b_k \le B(D_k)$, combine (7.1) with Lemma 7.1. The top-rank contribution is $< A_k$. For $r = k - s$, note

$$\binom{N_{k-s}-1}{k-s-2} \le \binom{N_k-1}{k-2}.$$

Lemmas 7.2 and 7.3 show that the ratio of the sum of all lower-rank contributions to A_k is bounded by

$$4 \sum_{s \ge 1} 2^{-s(s+5)/2} < 0.533 < 1.$$

Hence

$$B(D_k) < 2A_k = 2^{N_k+2} \binom{N_k-1}{k-2},$$

and $b_k \le B(D_k)$ gives the result. □

8 Point-count bound for the lower-rank boundary

8.1 Clearing denominators

In ordinary projective coordinates $x = (x_1, \dots, x_r)$, define

$$\Delta_r(x) = \prod_{0 \ne v \in \mathbb{F}_2^r} (v \cdot x), \qquad P_r(x) = \Delta_r(x) \ell_r(x).$$

Lemma 8.1. *P_r is a nonzero homogeneous polynomial of degree $2^r - 2$ over $\mathbb{F}_2$, and*

$$X_r \subset V(P_r) \subset \mathbb{P}^{r-1}.$$

Proof. Each summand of ℓ_r has denominator one nonzero linear form. Multiplication by Δ_r leaves a product of the remaining $2^r - 2$ linear forms. Nonvanishing follows from Lemma 3.1 in rank r, equivalently from the explicit decomposition

$$\ell_r(x', T) = \ell_{r-1}(x') + C/L(T).$$

□

Lemma 8.2. *For $q = 2^n$,*

$$\#X_r(\mathbb{F}_q) < \frac{2^r q^{r-1}}{q-1}.$$

Proof. The affine cone $V(P_r) \subset \mathbb{A}^r$ is defined by a nonzero polynomial of degree $2^r - 2$. Schwartz–Zippel gives fewer than $2^r q^{r-1}$ affine zeros. Every projective point contributes exactly $q - 1$ nonzero points to the cone. □

By Proposition 5.1,

$$D_k \setminus X_k = \bigsqcup_{r=2}^{k-1} \bigsqcup_{\dim W = r} X_r.$$

Theorem 8.3. *If $k \geq 17$ and $n \geq 2k + 1$, with $q = 2^n$, then*

$$\frac{\#(D_k \setminus X_k)(\mathbb{F}_q)}{q^{k-2}} < \frac{1}{4} + 2^{-k-5}.$$

Proof. By Lemma 8.2, Gaussian symmetry, and $s = k - r$,

$$\frac{\#\text{boundary}}{q^{k-2}} < \frac{q}{q-1} \sum_{s=1}^{k-2} \begin{bmatrix} k \\ s \end{bmatrix}_2 2^{k-s} q^{-s}.$$

For $s = 1$,

$$\begin{bmatrix} k \\ 1 \end{bmatrix}_2 = 2^k - 1,$$

and $q \geq 2^{2k+1}$ gives

$$\frac{(2^k - 1)2^{k-1}}{q-1} < \frac{1}{4}.$$

For $s \geq 2$, Lemma 7.2 gives the s-th summand, apart from $q/(q-1)$, as

$$< 2^{2+k-s(k+s+2)}.$$

The first term, $s = 2$, is 2^{-k-6}, and successive terms decrease by at least $2^{-(k+7)}$. Since $q/(q-1) < 1 + 2^{-35}$, the entire tail is $< 2^{-k-5}$. □

9 Positivity in the large-parameter range

Put $m = k - 2$. Corollary 6.4, Lemma 6.5, Theorem 7.4, and the boundary decomposition give

$$\#X_k(\mathbb{F}_q) \geq q^m - \beta_k q^{m/2} - \#\text{boundary}. \tag{9.1}$$

We deliberately discard the positive terms $1 + q + \cdots + q^{m-1}$.

9.1 The case $k = 17$

Proposition 9.1. *If $k = 17$ and $n \geq 35$, then $X_{17}(\mathbb{F}_{2^n}) \neq \varnothing$.*

Proof. Here $N_{17} = 184$ and

$$\binom{183}{15} < 2^{72},$$

so $\beta_{17} < 2^{258}$. With $n \geq 35$ and $m = 15$,

$$q^{m/2} \geq 2^{262.5},$$

hence $\beta_{17}/q^{m/2} < 2^{-4.5} < 1/16$. Theorem 8.3 gives a boundary ratio $< 1/4 + 2^{-22}$. Thus the normalized right-hand side of (9.1) is positive. □

9.2 The cases $k \geq 18$

Proposition 9.2. *If $k \geq 18$ and $n \geq 2k + 1$, then $X_k(\mathbb{F}_{2^n}) \neq \varnothing$.*

Proof. Use

$$\binom{M}{r} \leq \left(\frac{eM}{r}\right)^r.$$

Since

$$\frac{N_k - 1}{k-2} = \frac{k+7}{2} + \frac{3}{k-2},$$

we have $e(N_k - 1)/(k-2) < 2k$ for $k \geq 18$, so

$$\binom{N_k - 1}{k-2} < (2k)^{k-2}.$$

Also $\log_2 k \leq k/4$ for $k \geq 18$. At the worst value $n = 2k + 1$,

$$\log_2\left(\frac{\beta_k}{q^{(k-2)/2}}\right) \leq -\frac{k^2}{4} + \frac{9}{2}k - 2.$$

The right-hand side equals -2 at $k = 18$ and decreases strictly thereafter. Hence the middle ratio is at most $1/4$. By Theorem 8.3, the boundary ratio is $< 1/4 + 2^{-k-5}$. Therefore the total error ratio is < 1. □

9.3 The threshold $k = 16$

Proposition 9.3. *If $k = 16$ and $n \geq 37$, then $X_{16}(\mathbb{F}_{2^n}) \neq \varnothing$.*

Proof. Here $N_{16} = 165$ and

$$\binom{164}{14} < 2^{66},$$

so $\beta_{16} < 2^{233}$. With $m = 14$ and $n \geq 37$,

$$q^{m/2} = q^7 \geq 2^{259},$$

so the middle ratio is $< 2^{-26}$. The same boundary estimate is far below $1/4$ at $n = 37$, and the total normalized error is < 1. □

10 Finite tail and the main proof candidate

10.1 Published input

We use the following published results:

- Hou–Zhao [4, Thm. 5.4]: for $n \geq 8$, the lower half is covered at least through $3 \leq k \leq \lceil n/3 \rceil + 2$, with the complementary upper range; Corollary 5.5 gives the uniform range $k \leq 12$, and Corollary 5.6 settles all odd $n \leq 27$.
- Hou–Zhao [5]: Carlet's conjecture holds whenever n is not prime.
- Carlet–Hou [3, Table 2]: $15, 16 \in K_{31}$.
- Complementary-dimension symmetry: $k \in K_n \iff n - k \in K_n$.

10.2 Bookkeeping for odd primes

Restrict to odd prime n and $3 \leq k \leq (n-1)/2$. If $n \leq 27$, use HZ1. For $n = 29$, $\lceil 29/3 \rceil + 2 = 12$, leaving only $k = 13, 14$, settled by Appendix A. For $n = 31$, the threshold is 13, so among $k \leq 15$ only $k = 14, 15$ remain; $k = 14$ is settled by Appendix A and $k = 15$ by CH Table 2. For $n \geq 37$, $\lceil n/3 \rceil + 2 \geq 15$, so $k \leq 15$ is covered by HZ1; $k = 16$ by Proposition 9.3; and $k \geq 17$ by Propositions 9.1 and 9.2 because the lower-half condition implies $n \geq 2k + 1$.

Main Proof Candidate 10.1. *For every odd n and every $3 \leq k \leq n-3$, there exists a k-dimensional $\mathbb{F}_2$-subspace $E \subset \mathbb{F}_{2^n}$ such that*

$$\sum_{0 \neq x \in E} x^{-1} = 0.$$

Status: complete proof candidate; independent audit pending.

Proof. If n is composite, use HZ2. Now suppose n is an odd prime. By complementary-dimension symmetry it is enough to treat $3 \leq k \leq (n-1)/2$. For $n \leq 27$, use HZ1. The cases $n = 29$ and $n = 31$ are covered by Appendix A together with CH and HZ1. If $n \geq 37$, then $k \leq 15$ is covered by HZ1, $k = 16$ by Proposition 9.3, and $k \geq 17$ by Propositions 9.1 and 9.2. Proposition 3.2 converts a rational point on X_k to the desired zero-sum subspace. Complementary symmetry gives the upper half. □

Logical status

If the new/combined arguments in Theorem 4.2, Theorem 5.5, Theorems 6.2–6.3, Theorem 7.4, and Theorem 8.3 are confirmed by independent review, Candidate 10.1 becomes a theorem. The purpose of this manuscript is to expose those nodes in a form suitable for line-by-line verification.

11 Audit points and reproducibility

11.1 Genuine issues already found and repaired

- An earlier draft incorrectly claimed $H^i(D_k) \cong H^i(Q_k)$ above the middle degree. The present argument uses weak Lefschetz below the middle and Verdier duality on D_k above it.
- A web-text extraction initially obscured the full statement of EPW Proposition 2.14. The published statement does include $P_M(t) = 1 \iff L(M)$ modular.

- The coarse β_{16} bound fails at the formal threshold $n = 33$, but 33 is composite and is covered by HZ2. The first remaining prime is 37, where the middle error is $< 2^{-26}$.
- The coordinate-ring interface between the EPW reciprocal plane and the Pink–Schieder reciprocal ring has been checked: in the arrangement used here, both are $\operatorname{Spec} R_V$.
- The quadratic model has been upgraded from a pointwise finite-field correspondence to a scheme-level construction over an arbitrary base ring, including all unit conditions.

11.2 Reproducible supplementary material

The script `verify_carlet_paper.py` implements $\mathbb{F}_{2^n}$ using exact integer bit operations and checks Rabin irreducibility, binary rank, the complete subspace-polynomial recurrence, exhaustive reciprocal sums over 8191/16383 nonzero elements, and the exact $k = 16, 17$ numerical thresholds. No floating-point arithmetic is used in the certificate verification. The theoretical argument itself does not depend on the code.

11.3 Suggested independent-review checklist

- Check that the reverse morphism in Theorem 4.2 factors through Ω_r over an arbitrary base ring.
- Check the exact identification of the normal factor in Pink–Schieder Proposition 8.3 with the EPW reciprocal arrangement used in Lemma 5.2.
- Check the smooth slicing and intersection-complex shifts in Theorem 5.5.
- Check the perverse/cosupport convention in Wan–Zhang Lemma 2.3.1 and the degree shift in Theorem 6.2.
- Check the rank-filtration spectral sequence and every combinatorial inequality in Theorem 7.4.
- Check the projective cone estimate and rank multiplicities in Theorem 8.3.
- Run the supplied verifier independently and compare every hexadecimal datum in Appendix A.

A Exact certificates for $n = 29$ and $n = 31$

Field elements are encoded as bitmasks in the polynomial basis: $\sum a_i \alpha^i$ is encoded by the integer $\sum a_i 2^i$, so addition is XOR.

A.1 $n = 29$, $k = 13$

Work in

$$\mathbb{F}_{2^{29}} = \mathbb{F}_2[\alpha]/(\alpha^{29} + \alpha^2 + 1).$$

A basis is

```
0x00000001
0x0236c002
0x17b15004
0x16f41008
```

```
0x0aa45010
0x17b08020
0x178f8040
0x152d9080
0x0b024100
0x1ecb9200
0x11e0c400
0x092bd800
0x0561e000
```

For $L_E(X) = \sum_{i=0}^{13} c_i X^{2^i}$, the coefficients $c_0, \dots, c_{13}$ are

```
0x1a8d2e2f
0x00000000
0x0b39aa9c
0x0f919f54
0x06a764cc
0x0d93d023
0x0cfe406e
0x02d45aa5
0x03f76408
0x1bc649a2
0x10330d40
0x191ffaf2
0x0a266fda
0x00000001
```

Binary rank $= 13$, $c_1 = 0$, and the exhaustive reciprocal sum over all 8191 nonzero elements is 0.

A.2 $n = 29$, $k = 14$

In the same field, take

```
0x00000001
0x00538402
0x1abd8404
0x18eb0408
0x1aec0010
0x0d550020
0x15c60040
0x1f908480
0x14320500
0x04958200
0x13470800
0x0f4a1000
0x0f172000
0x14fe4000
```

The coefficients $d_0, \dots, d_{14}$ are

```
0x19ce08e7
0x00000000
0x08d8433a
0x04bd1933
0x121c9c05
0x1b5ac54f
0x046f1aa3
0x1b021f2d
0x1674a568
0x0eee40d5
0x037b879d
0x010d3c06
0x1b2e2c21
0x02427c2c
0x00000001
```

Binary rank $= 14$, $d_1 = 0$, and the exhaustive reciprocal sum over all 16383 nonzero elements is 0.

A.3 $n = 31$, $k = 14$

Work in

$$\mathbb{F}_{2^{31}} = \mathbb{F}_2[\beta]/(\beta^{31} + \beta^3 + 1).$$

A basis is

```
0x00000001
0x3b088937
0x171012ec
0x6afce627
0x5e288246
0x4d96a9a8
0x4bf132d0
0x59374007
0x39117efd
0x460fc715
0x1beb9677
0x566e2dcd
0x19437aab
0x00058364
```

The subspace-polynomial coefficients are

```
0x1b99e28a
0x00000000
0x435fe9c9
0x04ecca32
0x148e3fe3
0x79802dca
0x32fcb2ff
0x0019ea0d
```

```
0x3038b402
0x7f441af6
0x18ecd503
0x347b8d50
0x285c194d
0x48766441
0x00000001
```

Binary rank = 14, the X^2 coefficient is 0, and the exhaustive reciprocal sum over all 16383 nonzero elements is 0.

B Verification algorithm

B.1 Subspace-polynomial verifier

Starting from $L_{\{0\}}(X) = X$, the program inserts the basis vectors one at a time using

$$L^+(X) = L(X)^2 + L(v)L(X).$$

The final coefficient list is compared with Appendix A and the X^2 coefficient is checked to be zero.

B.2 Exhaustive reciprocal sum

The program enumerates all nonzero vectors in each span by Gray code. Each nonzero x is inverted as x^{2^n-2} and accumulated by XOR. This route is logically independent of Lemma 2.2.

B.3 Rabin irreducibility

Because 29 and 31 are prime, Rabin's criterion reduces here to

$$X^{2^n} \equiv X \pmod{p}, \qquad \gcd(p, X^2 - X) = 1.$$

The supplementary program checks these identities exactly.

C Numerical thresholds

k	N_k	$\log_2 \binom{N_k-1}{k-2}$	use
16	165	65.8390	$n \geq 37$ separately
17	184	71.6348	gateway $n \geq 35$
18	204	77.5198	start of analytic range

The exact thresholds used above are

$$\binom{183}{15} < 2^{72}, \qquad \binom{164}{14} < 2^{66}.$$

For $k \geq 18$, the exponent

$$-\frac{k^2}{4} + \frac{9}{2}k - 2$$

equals -2 at $k = 18$ and decreases strictly thereafter. The supplementary script checks the half-integer $k = 17$ threshold by exact integer squaring.

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SUPPLEMENT: THE EVEN-DEGREE CASE OF CARLET'S CONJECTURE

WANG QUANZHI

Relation to the accompanying proof candidate. This supplement proves the even-degree statement by a finite-field counting and subspace-polynomial argument. It is a separate route from the reciprocal-geometric argument proposed for odd degree in the accompanying manuscript. The odd-degree argument remains a proof candidate requiring independent specialist review; the existence of this even-degree proof does not certify that argument.

1. Statement and coefficient criterion

Let K be a finite field of characteristic 2 and order $q = 2^n$. For an $\mathbb{F}_2$-subspace $E \subseteq K$, put

$$L_E(T) = \prod_{x \in E} (T + x) = T^{2^d} + c_{d-1} T^{2^{d-1}} + \cdots + c_1 T^2 + c_0 T, \qquad d = \dim_{\mathbb{F}_2} E.$$

The reciprocal-sum criterion says

$$\sum_{x \in E \setminus \{0\}} x^{-1} = 0 \quad \Longleftrightarrow \quad c_1 = 0.$$

Indeed, the coefficient of T^2 in $L_E(T)$ is the product of the nonzero elements of E times the reciprocal sum; that product is nonzero.

Theorem 1. *If $n \geq 4$ is even, then for every integer k with $2 \leq k \leq n-2$, there is a k-dimensional $\mathbb{F}_2$-subspace $E \subseteq K$ such that $\sum_{x \in E \setminus \{0\}} x^{-1} = 0$.*

2. Base dimensions two and three

For even n, the field K contains its unique subfield $\mathbb{F}_4 = \{0, 1, \omega, \omega^2\}$, where $\omega^2 + \omega + 1 = 0$. Then

$$\sum_{x \in \mathbb{F}_4^\times} x^{-1} = 1 + \omega + \omega^2 = 0,$$

so dimension two is settled.

Now suppose that $n \geq 6$ is even. Choose uniformly among the three-dimensional subspaces containing 1, and let $p_3(0)$ be the proportion for which the T^2 coefficient of L_E is zero. Write $\mathrm{Tr} : K \to \mathbb{F}_2$ for the field trace and

$$\mathrm{Kl}_n = \sum_{v \in K^\times} (-1)^{\mathrm{Tr}(v + v^{-1})}.$$

For later use, let $p_2(v)$ denote the proportion of two-dimensional subspaces containing 1 whose T^2 coefficient is v. The exact one-step subspace-extension count, with the even-degree trace

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term retained, is

$$(q-4)p_3(0) = 1 - \frac{2}{q-2} + 3\sum_{v\in K^\times} (-1)^{\mathrm{Tr}(v^{-1})} p_2(v), \tag{1}$$

$$(q-2)p_2(v) = 1 + (-1)^{\mathrm{Tr}(v)} - 2\mathbf{1}_{v=1} \qquad (v \neq 0). \tag{2}$$

Here $\mathbf{1}_{v=1}$ is 1 when $v = 1$ and 0 otherwise. For completeness, let $N_d(c)$ be the number of all d-subspaces whose T^2 coefficient is c, let $H_d(c)$ count those that also contain 1, and let A_d be the total number of d-subspaces containing 1. The raw Artin–Schreier fiber count, obtained by applying the character identity for the fibers of $x \mapsto x^2 + x$ to each d-subspace, is

$$2^d H_{d+1}(b) = N_d(b+1) + \sum_{w\neq 0} (-1)^{\mathrm{Tr}(w)} H_d((b+1)/w), \qquad b \neq 1.$$

Incidence counting gives $(2^d - 1)N_d(c) + H_d(0) = A_d$ for $c \neq 0$, and counting adjacent dimensions gives $A_{d+1}/A_d = (q - 2^d)/(2^d(2^d - 1))$. Dividing the raw identities by the appropriate totals yields (1) at $d = 2, b = 0$. In (2), the two-dimensional coefficient count is explicit; the correction at $v = 1$ removes the repeated line, and $\mathrm{Tr}(1) = 0$ because n is even.

Multiply (2) by $(-1)^{\mathrm{Tr}(v^{-1})}$ and sum over $v \neq 0$. The nontrivial additive-character sum over $K^\times$ is -1, the product of the two characters is $(-1)^{\mathrm{Tr}(v+v^{-1})}$, and the exceptional $v = 1$ term contributes -2. Consequently

$$(q-2)\sum_{v\in K^\times} (-1)^{\mathrm{Tr}(v^{-1})} p_2(v) = \mathrm{Kl}_n - 3.$$

Substituting this into (1) and simplifying gives the exact identity

$$(q-2)(q-4)p_3(0) = q - 13 + 3\mathrm{Kl}_n.$$

Consider the elliptic curve $C : y^2 + xy = x^3 + 1$. Its discriminant is 1, so it is nonsingular in characteristic 2. For each $x \neq 0$, the equation in y has $1 + (-1)^{\mathrm{Tr}(x+x^{-1})}$ solutions; at $x = 0$ it has one solution, and the projective point at infinity contributes one more. Hence

$$\#C(K) = q + 1 + \mathrm{Kl}_n.$$

The Hasse bound for elliptic curves (see [1]) yields $|\mathrm{Kl}_n| \leq 2\sqrt{q}$. Since $q \geq 64$,

$$q - 13 + 3\mathrm{Kl}_n \geq q - 13 - 6\sqrt{q} = s^2 - 6s - 13 \geq 3 > 0, \qquad s = \sqrt{q} \geq 8.$$

Also $(q-2)(q-4) > 0$. Thus $p_3(0) > 0$, which proves existence of a three-dimensional zero-sum subspace for every even $n \geq 6$. When $n = 4$, the allowed range contains only $k = 2$, already handled above.

3. Dimension extension

Lemma 1. *If zero-sum subspaces of dimensions a and b exist in K and $ab < n$, then a zero-sum subspace of dimension $a + b$ exists.*

Proof. Let E, F be such subspaces and let $I = L_E(K)$. The map $L_E : K \to K$ is $\mathbb{F}_2$-linear with kernel E, so K/I has dimension a. Consider the linear map

$$K \longrightarrow \mathrm{Hom}_{\mathbb{F}_2}(F, K/I), \qquad \lambda \longmapsto (x \mapsto \lambda x \bmod I).$$

Its target has dimension $ab < n$, so its kernel contains a nonzero λ. Hence $\lambda F \subseteq I$. Put $G = L_E^{-1}(\lambda F)$. The kernel-image dimension formula gives $\dim G = a + b$, and the root sets

give $L_G = L_{\lambda F} \circ L_E$. Scaling preserves the vanishing of the T^2 coefficient. For linearized polynomials A, B, the T^2 coefficient of $A \circ B$ is

$$A'(0)\,[T^2]B + [T^2]A\,([T]B)^2.$$

Both terms vanish here because E and λF are zero-sum. Therefore G is zero-sum. □

4. Complementary dimensions and conclusion

For any E, set $I = L_E(K)$. Since L_E has kernel E, $\dim I = n - \dim E$. Moreover,

$$L_I \circ L_E = T^q - T,$$

because both sides are monic degree-q polynomials with every element of K as a root. Comparing the T^2 coefficients shows that, for $q > 2$, E is zero-sum if and only if I is zero-sum. Thus existence in dimension k is equivalent to existence in dimension $n - k$.

We prove existence first for $2 \le d \le n/2$. Dimensions two and three are the base cases. For $d \ge 4$, apply the dimension-extension lemma to dimensions $d-2$ and 2. Its hypothesis is $2(d-2) < n$, which follows from $d \le n/2$. Induction proves the claim through the lower half. For any $2 \le k \le n-2$, use this result directly if $k \le n/2$; otherwise $2 \le n - k < n/2$, and the complementary-dimension equivalence supplies dimension k. This proves the theorem for every finite field of order 2^n, without restricting to a particular field model.

References

[1] J. H. Silverman, *The Arithmetic of Elliptic Curves*, second edition, Graduate Texts in Mathematics 106, Springer, 2009, Chapter V, Theorem 1.1.